MTU_Value*Service*
Technical Documentation

Diesel Engine
8 V 4000 R41
8 V 4000 R41L
8 V 4000 R41R

Instructions for Exchange
of Sub-assemblies
MS22029/00E

mtu

Printed in Germany

This handbook is provided for use by maintenance and operating personnel in order to avoid malfunctions or damage during operation.
Subject to alterations and amendments.

1 General Information

1.1 Important Notes

1.1.1 Prerequisites for maintenance work and general assembly instructions

This maintenance manual is intended for the technical personnel assigned to exchange of assemblies.

Contents: Instructions on exchange of assemblies

Prerequisites for maintenance work

If maintenance work is carried out by the customer, the following needs to be ensured:

- Observance of safety regulations.
- Deployment of trained and qualified personnel.
- Suitable shop equipment and general tools.
- Suitable testing equipment.
- Approved special tools.

General assembly instructions

Always use new sealing and securing elements for assembly.

Components that come into contact with oil, fuel, engine coolant and combustion air must be clean. Prior to assembly, clean components with “Special cleanness” requirement (e.g. oil and fuel-carrying parts) using suitable cleaning methods and then check for cleanness. Remove any packaging from components immediately before assembly.

Do not wash elastomer components (e.g. rubber, etc.) using diesel fuel, solvent or cold cleaner.

- Remove any oil or fuel film. Wipe parts down with a dry cloth to clean them.
- Do not paint elastomer components, such as e.g. engine mounts, damping elements, clutches and V-belts. Only install them after painting the engine or cover them prior to painting.

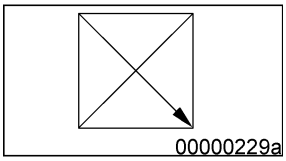
Radial shaft seals that have been treated with oil by the manufacturer, have a defined moisture expansion on delivery. Prior to installation, they may only be cleaned using abrasion-proof paper (not washed).

If not otherwise specified, the surfaces of parts that slide on each other should be moistened with SAE 30 engine oil.

If not otherwise specified, O-rings and surfaces (bores or shafts) that slide along them during installation, should be coated with grease. During the assembly of O-rings with counterrings in coolant pumps, observe the assembly instructions. After assembling O-rings in the grooves on shafts, push a rounded scriber under the O-ring in circumferential direction, if the O-ring diameter is large enough. The O-ring may not be damaged.

Prior to assembling shaft seals:

- Coat shaft and sealing lip of shaft seal with grease and coat shaft running surface with thin-film lubricant or SAE 30 engine oil.
- If no other information is specified on the drawing, coat the outer surface of mounting bores on metal outer liners with surface sealant. Coat the outer surface of elastomer outer liners or combined metal/elastomer outer liners with ethanol.



Graphical symbol applies to radial shaft seals and depends on the position. Dimension arrow indicates the position of the sealing lip.

Use sealing paste to fix the position of flat gaskets. Apply sealing paste thinly and in the form of dots on the flat gasket or opposite surface. Flat gaskets should be attached to a component shortly after applying the sealing paste and the sealing point should then be bolted (at the latest after 20 minutes)

Prior to installing antifriction bearings, lubricate the bearing seats. Only take bearings out of their original packaging immediately before installation. Do not remove corrosion inhibitor from bearings in original packaging. To clean the antifriction bearings, use denatured spirit or acid-free kerosene. After cleaning, lubricate bearings with SAE 30 engine oil.

- Do not apply assembly forces (axial) over rolling elements and do not knock bearing rings with hammers (use assembly aids).
- Do not use naked flames to heat inner bearing races!
- The temperature should be between 80°C and 100°C and may not exceed 120°C.
- Freezing as a method for the assembly of anti-friction bearings is not permitted (danger of cracks, rust formation due to condensate).

Do not lubricate dry bearings with oil.

When installing gears, use SAE 30 engine oil to moisten teeth.

Prior to assembly, mating and contact surfaces of components (e.g. contact surfaces for centering, flange and sealing faces, joint surfaces of interference fit assembles) must be clean, bright or have the specified surface protection and also be free of surface irregularities and damage. Remove any corrosion inhibitor (e.g. oil, grease) from mating and contact surfaces.

After joining components that are chilled with liquid nitrogen, remove any condensate and preserve with SAE 30 engine oil.

Prior to installation in immersion sleeves, coat the sensor area of measuring sensors with long-life lubricant.

Preassemble line links with cutting ring union in the bench vise and tighten; coat thread with thin-film lubricant first.

If components need to be marked by means of etching, use neutralization agent to remove the marking solution afterwards. Preserve the relevant areas with SAE 30 engine oil.

The mating/contact surfaces of components used in the hot-component area (e.g. V-belt clamps, bellows, plug-in pipes), should be coated with assembly paste, if not otherwise specified.

The assembly surfaces of screws, nuts, washer components and parts to be tensioned must be clean, bright or have surface protection and also be free of surface irregularities and damage. Remove any corrosion inhibitors (e.g. oil, grease). Prior to assembly according to tightening specifications, coat threads and screw bearing surfaces with lubricant.

If not otherwise specified, use SAE 30 engine oil as lubricant and in the hot-component area assembly paste.

Screw connections without tightening specification can be tightened freely, i.e. machine-tightening with screwdriver or hand-tightening with open-end wrench or box wrench. For machine-tightening, take the torque specification for thread size and property class from the general tightening specifications.

Screw connections with tightening specification:

- Establish the screw connections by hand-tightening with snap wrench or torque indicator wrench. Apply tightening torque without taking the tolerance specified on the snap wrench into account. If a torque indicator wrench is used, the torque must be within the tightening torque limit range. Proceed in the same way for connections protected against torsion. This information also applies to test torques.
- If no tolerance is specified for the tightening torque, the tightening tolerance is +10 % of the specified torque.
- Rotation angle tightening:
  The angles of further rotation specified in the torque specification must be established. They may be exceeded within the specified tolerance range. If no tightening torque is specified, maintain the following tolerances: +5 ° for angle of further rotation below/equal 90 ° +10 ° for angle of further rotation above 90 °. Prior to rotation angle tightening, apply a color line mark to each screw head to allow the inspection of the correct angle of rotation after tightening (exception: the color line mark is not required for tightening using self-monitoring NC screwdrivers).
- Elongation tightening:
  Tighten according to tightening specification, taking the tightening tolerance into account.

TIM ID: 0000020679 – 003

